The $k = 4$ Macaulay solve, 6× faster in wall time, the same arithmetic

2026-10-13. Commit `029309579`, `src/cryptanalysis/gaudry_quartic.rs`; study `research/gaudry_quartic_speed_20261013/`. **Class: engineering** — the counted cost C_4 is unchanged, so the crossover with rho does not move.

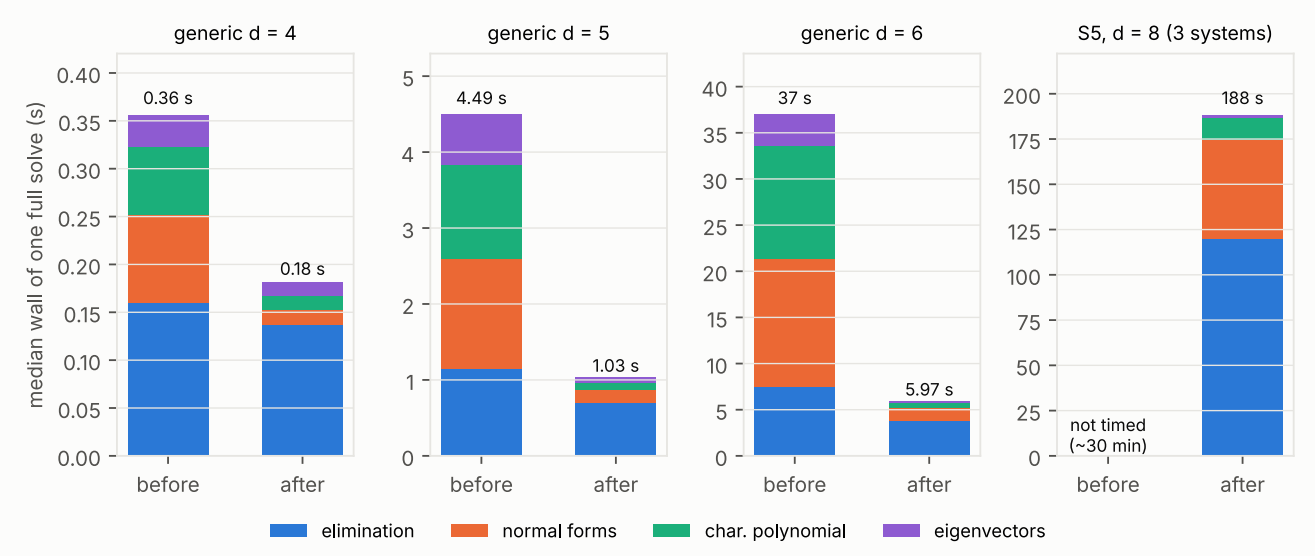
Change

Characteristic polynomial and eigenvectors moved from `u128` remainders, one entry at a time, to `u16` rows with the vectorised `axpy`, rows in parallel (the Hessenberg step's elementary similarities commute, so its row operations precede its column operations with the same result). Normal forms resolved in parallel blocks of 128 columns. `axpy` compiled for AVX2/AVX-512 and chosen at run time. Same operations, counts and outputs: identical on the committed $d = 3…5$ panel, pinned by a unit test, equal in `total_muls` on all 72 solves here, and $1.2269 \cdot 10^{12}$ on the S_5 solve, §11.17's C_4.

Result

Interleaved before/after, generic planted systems at $p = 269$, 12 solves per arm and degree, median wall time in ms.

d	dim	elimination	normal forms	char. poly.	eigenvectors	whole solve
4	256	155 ⟶ 128	91 ⟶ 15	70 ⟶ 17	33 ⟶ 14	354 ⟶ 184 (**1.9**×)
5	625	1,074 ⟶ 690	1,456 ⟶ 179	1,192 ⟶ 93	608 ⟶ 73	4,368 ⟶ 1,067 (**4.1**×)
6	1,296	7,116 ⟶ 3,682	13,961 ⟶ 1,412	12,342 ⟶ 507	3,364 ⟶ 169	37,433 ⟶ 5,862 (**6.4**×)



Median wall time of one full solve by phase. S_5 ($d = 8$, dimension 4,096, three systems) after only: 164–203 s a solve; before not timed here (about 30 min a solve in earlier runs on this host).

The gain grows with size; the elimination is now about 64 % of the S_5 solve and the next lever.

Environment: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU with AVX-512, 15 GiB RAM, rustc 1.97.0; shared host, otherwise idle; four rayon threads. Conductor control plane unreachable: no task card.